

Fats



- Fats are macronutrients that are required by the body in appropriate amounts. They are classified as healthy “good fats” and unhealthy “bad fats” for general term use.
- Healthy fats are required by the body for proper functioning.
- Fats have high calorific value. Per gram of fat provides 9Kcal, more energy per gram compared to protein and carbohydrates.

Dietary fats

Saturated fatty acid: Unhealthy fats

Saturated fats are considered “unhealthy fats” since they increase LDL cholesterol “bad cholesterol” and increase the risk of getting heart diseases.

Found naturally in many foods, especially

animal foods such as ghee, butter, meat and dairy products

plant sources such as coconut oil, palm and palm kernel

They are also found in deep-fried foods and packaged food items.



Trans fats are formed during food processing to turn liquid oils into solid fats.

Trans fatty acids (TFA) not only increases the levels of LDL cholesterol and cholesterol which are bad for the heart but also decreases levels of good (HDL) cholesterol. Therefore it is best to avoid trans fatty acids.

Found in fried foods, commercial baked goods (donuts, cookies, crackers), processed foods, and margarines (vanaspati).

Look for the terms 'partially hydrogenated' or 'shortening' on food labels as these are the sources of trans fatty acids.



Unsaturated fatty acids: Healthy fats

Unsaturated fats are termed as "healthy fats" as they lower the risk of heart diseases. From the total calories coming from fats, it should be primarily from unsaturated fatty acids, in it less than 10% from SFA and less than 1% from TFA.

Sources include: Sunflower, corn, soybean, canola and flaxseed oils; walnuts, flax seeds and fish.

It is further divided into: **Monounsaturated fatty acids** and **Polyunsaturated fatty acids**.

Monounsaturated fatty acids (MUFA):

MUFA are primarily derived from plants. Examples include olive oil, peanut oil, sesame oil, avocados and nuts.



Polyunsaturated fatty acids (PUFA):

PUFA are called essential fatty acids. Essential fatty acids – are not synthesized in the body therefore the requirements have to be met from diet. Vegetable oil (Sunflower, corn, soyabean, canola and flaxseed), walnuts, flaxseeds and fish.



Alpha-linolenic acid (omega-3 or n-3):

The sources of omega-3 fatty acids includes fish and other seafood, particularly cold-water fatty fish like salmon, mackerel, tuna, herring, and sardines. Good sources from plant foods are flaxseeds, walnuts, chia seeds and canola oil.



Linoleic acid (omega-6 or n-6):

Good sources of omega-6 fatty acids include safflower oil, sunflower oil, corn oil, soybean oil, and sunflower seeds, walnuts, pumpkin seeds.



What are the functions of fats?



They are required for transportation and absorption of fat-soluble vitamins (A, D, E and K) in the body.



They increase the palatability and satiety of the food.



Fats are essential for the development of the brain and eyes in infants and young children.



It is crucial for maintaining good skin in both children and adults.



It is an essential component of all cell membranes.



Fat tissue supports and cushions organs and nerves in your body, shielding them from harm due to impacts or shocks.



They provide insulation and keeps body warm.

What is difference between visible and invisible fats?

Invisible fats: Invisible fat are fats present as integral components of food. These fats are also known as hidden fats.

Visible fats: Visible fats are those which are used while cooking such as Vegetable oil, ghee, butter and Vanaspati.

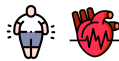
What will happen when there is too low fat intake in the diet?

Inadequate dietary fat, along with insufficient protein and carbohydrates, can result in weight loss and hinder growth. Deficiencies in omega-3 and omega-6 PUFAs may cause neurological issues, skin problems, and stunted growth, primarily in those severely malnourished or with conditions like cystic fibrosis, pancreatic issues, or intestinal damage, impairing fat absorption.



What happens when there is overconsumption of fats?

Consuming too much fat in the diet can increase the chance of becoming obese, developing heart problems, and certain cancers. Increased intake of lots of saturated fats is known to raise the risk of heart disease.



How much fat should be included in the diet per day?

ICMR-NIN recommends the upper limit for visible fat intake based on individual physical activity.



For adults with sedentary, moderate, and heavy exercise, the recommended dietary fat intake has been set at 25, 30, and 40 g/d for men and 20, 25, and 30 g/d for women.

Fats are insoluble in water unlike carbohydrates and proteins. Hence, they are absorbed and transported through lymphatic system.

Bile acid, secreted from gall bladder, is required for the emulsification (break down) of fats.

Fats are digested into small intestine wherein they are converted into long chain fatty acids (LCFAs) and then to short chain fatty acids (SCFAs) and finally into digested lipid products which are ready for absorption.

Fats form complexes with proteins (lipoproteins) for transportation.

Tip: Cholesterol is found only in animal source foods- plant based oils are naturally cholesterol free! However, it is needed in smaller quantities as it plays key role in membrane integrity, hormones synthesis and bile salts that in turn help in metabolism.

Nutrition label:

Be cautious about food items labelled as “zero-trans fat” or “trans-fat free”. Such food items are usually high in saturated fats. Check the labels properly before making any decision.

Nutrition Facts		Nutrition Facts	
		6 servings per container Serving size 1 cup (230g)	
		Calories 245	
% Daily Value*		% Daily Value*	
Total Fat 12g	14%	Total Fat 12g	14%
Saturated Fat 2g	10%	Saturated Fat 2g	10%
Trans Fat 0g		Trans Fat 0g	
Cholesterol 8mg	3%	Cholesterol 8mg	3%
Total Sugars 10g		Total Sugars 10g	
Includes 4g Added Sugars		Includes 4g Added Sugars	
Protein 1g		Protein 1g	
*The % Daily Value (DV) tells you how much a nutrient in a serving of food contributes to a daily diet. 2,000 calories a day is used for general nutrition advice.		*The % Daily Value (DV) tells you how much a nutrient in a serving of food contributes to a daily diet. 2,000 calories a day is used for general nutrition advice.	

Fatty acid composition of Indian cooking oil*

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WHO suggested a fatty acid ratio of 1:1.5:1 for saturated : monounsaturated : polyunsaturated fatty acids and 1:5–10 alpha-linolenic acid (omega-3): linoleic acid (omega-6) in the food intake.

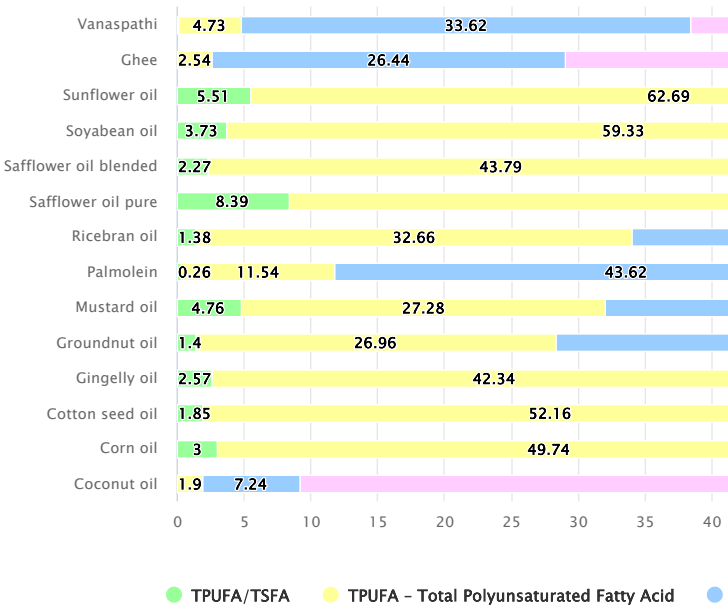
Since no one oil has the fatty acid profile in the correct ratio, it is advised to use combination of two or three oil in daily cooking or rotate the oil every three months.

Table of fatty acid composition of Indian cooking oil*

Edible oil	TSFA	TMUFA	TPUFA	TPUFA/TSFA
Coconut oil	90.84	7.24	1.90	0.02
Corn oil	16.60	33.67	49.74	3.00
Cotton seed oil	28.17	19.66	52.16	1.85
Gingelly oil	16.45	41.21	42.34	2.57
Groundnut oil	19.27	53.77	26.96	1.40
Mustard oil	5.73	66.98	27.28	4.76
Palmolein oil	44.84	43.62	11.54	0.26
Rice bran	23.63	43.71	32.66	1.38
Safflower oil pure	9.19	14.04	76.78	8.39
Safflower oil blended	19.31	36.90	43.79	2.27
Soyabean oil	15.90	24.77	59.33	3.73
Sunflower oil	11.39	25.92	62.69	5.51
Ghee	71.02	26.44	2.54	0.04
Vanaspathi	61.65	33.62	4.73	0.08

TSFA – Total Saturated Fatty Acid
TMUFA – Total Unsaturated Fatty Acid
TPUFA – Total Polyunsaturated Fatty Acid
* Dorni C, Sharma P, Saikia G, Longvah T. Fatty acid profile of edible oils and fats consumed in India.Food Chem. 2018;238:9-15.

Fatty Acid Com



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